Reconciling a multi-voice corpus by set-theoretic estimation: the "cause of the tides" run

Haiku extracted the frames; Lean machine-checked the verdict

thejeb project note

2026-07-24

Abstract

We ran the empirical pipeline of the companion plan end to end. Seven public-domain natural philosophers discussing one topic — the cause of the ocean tides — were extracted into a structured frame per author by **Claude Haiku 4.5** (open pass), unified into a canonical schema by **Claude Sonnet**, and re-extracted against that schema by Haiku; the resulting frames were encoded as STE property sets and the invariants of the disagreement were **machine-checked in Lean** against `Ste.Basic` (Combettes 1993). The verdict: the corpus is globally *inconsistent*; the obstruction is carried by the moon's role, which splits three ways (attraction / pressure / rejected); and the disagreement is *pervasive* rather than localized to one outlier — even the five-author attraction camp is internally inconsistent on the mechanism. This corrected the plan's armchair prediction of a single outlier (Galileo): the machine found *two* dissenters and a deeper coupling, which is the point of running it.

1 The run

The pipeline did not fix the schema up front. Three passes (prompts committed in `scripts/tides/`, outputs in `sources/tides/extraction/`):

1. **Haiku, open extraction.** Haiku read the seven passages and proposed its own (variable, value, quote) claims, over-splitting into ≈ 25 fine-grained variables (`pass1.json`).
2. **Sonnet, schema unification.** Sonnet merged the synonymous variables into a canonical **7-variable** schema with closed domains and a synonym map (`schema.json`): primary_cause, moon_role $\in$ {attraction, pressure, rejected}, mechanism (5 values), sun_role, spring_neap, earth_motion_required, quantitative.
3. **Haiku, schema-constrained re-extraction.** Haiku re-extracted every author against the canonical schema, emitting one assignment per author (value or silent) with a licensing quote (`frames.json`).

The extraction ran on Haiku 4.5 / Sonnet through the agent harness; the prompts are reusable and reproduce the frames.

2 The extracted frames

The canonical frames ($\cdot$ = silent):

Author	primary_cause	moon_role	mechanism	sun_role	quant.
Pliny	sun+moon	**attraction**	direct_corr.	reinforces	·
Bede	moon	**attraction**	direct_corr.	·	·
Kepler	moon	**attraction**	gravitational	·	·
Galileo	earth_motion	**rejected**	basin_motion	·	·
Descartes	sun+moon	**pressure**	vortex_pressure	reinforces	·
Newton	sun+moon	**attraction**	gravitational	weaker	math.
Laplace	sun+moon	**attraction**	dyn. oscill.	·	math.

3 The machine-checked invariants

Encoded in `Ste.TidesCorpus`: each author a is a property set `constraint` $a = \{f \mid f$ agrees with a's frame where
and the STE feasibility set is `feasibilitySet constraint` $= \bigcap_a$ `constraint` a. All of the following are proved (sorry-free):

- **Global inconsistency** (`feasible_eq_empty`): `feasibilitySet constraint` $= \varnothing$. No single reading of the tides satisfies all seven voices.
- **The three-way moon_role split gives three minimal cores**: any two authors in different camps already have disjoint property sets — `kepler_galileo_core` (attraction vs. rejected), `kepler_descartes_core` (attraction vs. pressure), `galileo_descartes_core` (rejected vs. pressure).
- **The disagreement is pervasive** (`pliny_kepler_mechanism_core`): even inside the five-author attraction camp, Pliny (direct correspondence) and Kepler (gravitational) are a minimal conflicting core on mechanism. So dropping the two moon_role dissenters does *not* restore consistency.
- **Positive structure on the most-contested variable** (`attraction_camp_moon_consistent`, `descartes_dissents`, `galileo_dissents`): on moon_role alone, the five attraction authors share a common reading, and exactly two voices dissent — Descartes (pressure) and Galileo (rejected).
- **The Combettes corollary** (`no_fair_reading`): because the formulation is inconsistent, every candidate estimand is unfair to some voice — the set-theoretic detection-of-invalid-information principle, instantiated on the corpus.

4 What the machine corrected

The companion plan predicted, from historical intuition, a *single* famous outlier (Galileo) and a nonempty consensus once he is dropped. The run refuted both halves:

1. There are **two** dissenters on the moon's role, not one: Galileo (rejected) *and* Descartes (pressure). The moon_role variable is a genuine three-way split.
2. Dropping the dissenters does **not** yield a consensus: the attraction camp is itself split on mechanism (direct correspondence vs. gravitational attraction vs. dynamical oscillation). The coupling is layered and pervasive.

This is exactly the value of a machine-checked empirical instance: the verdict is derived from the extracted frames, not from a plausible story, and it overturned the plausible story.

5 The boundary of trust

The Lean layer certifies the aggregation logic, not the reading of the text. Every asserted value in `frames.json` carries the source sentence that licenses it, so each frame is auditable against the passage; but a mis-extraction would yield a wrong-but-machine-checked verdict. The invariants above are conditional on the frames. The corpus passages are curated public-domain representative statements (provenance in `refs.bib`); pinning exact critical editions and expanding beyond seven voices is future corpus work.

6 Verified now vs. next

Mechanized: the feasibility verdict (empty), the minimal conflicting cores, the three-way split, the pervasive-coupling core, and the single-variable consensus/dissent structure, all tied to the verified `feasibilitySet`. Not yet mechanized for this instance: the *quantitative* invariants of the companion representation theory — the concept lattice of coherent voice-clusters (`Ste.HyperFrame`), the rectangle-cover complexity ρ and Čech obstruction magnitude (`Ste.RepresentationBounds`, `Ste.CechObstruction`), and the treewidth of the relation graph (`Ste.Treedecomp`). Those need the computable `Finset` refinement of the spec (plan phase P2) run on this instance; the qualitative verdict is complete.

References